

Student Name: Vikram Singh

Roll No: 12BIO004

Subject: Biology

School / Institution:	Amity International School, Noida
Student Name:	Vikram Singh
Roll Number:	12BIO004
Class:	XII-D
Subject:	Biology
Barcode:	BAR104
Date of Examination:	15 March 2025
Session:	2024-25

I hereby declare that the answers written in this booklet are my own. I have not used any unfair means during the examination.

Candidate's Signature: _____

Invigilator's Signature: _____

Room No: _____

SECTION A — Very Short Answer (1 mark each)

Q1. Define 'totipotency' in plants.

[1 mark]

Totipotency is the ability of a single plant cell to divide and differentiate to form a complete organism.

This forms the basis of tissue culture techniques in plant biotechnology.

Haberlandt (1902) first proposed this concept. A single meristematic cell can regenerate a whole plant.

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Q2. What is the significance of meiosis?

[1 mark]

Meiosis maintains the chromosome number constant across generations. It reduces diploid (2n) cells to haploid (n) gametes.

Meiosis also creates genetic variation through crossing over (recombination) during prophase I.

Without meiosis, chromosome number would double with each fertilisation.

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Q3. Name the enzyme that catalyses the dark reaction of photosynthesis.

[1 mark]

RuBisCO (Ribulose-1,5-bisphosphate carboxylase/oxygenase) catalyses the dark reaction (Calvin Cycle).

It is the most abundant enzyme on Earth.

Catalyses: $\text{CO}_2 + \text{RuBP (C}_5\text{)} \rightarrow 2 \text{ molecules of 3-PGA (C}_3\text{)}$ in the first step of carbon fixation.

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Q4. What are interferons?

[1 mark]

Interferons are glycoproteins (proteins) secreted by virus-infected cells.

They warn neighbouring cells to produce antiviral proteins that prevent viral replication.

They are species-specific but not virus-specific. Types: IFN- α , IFN- β (antiviral), IFN- γ (immune modulation).

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SECTION B — Short Answer (2 marks each)

Q5. Explain the process of double fertilisation in angiosperms.

[2 marks]

Double fertilisation is a unique feature of angiosperms (flowering plants).

Process:

1. Pollen grain germinates on stigma, pollen tube grows through style
2. Pollen tube reaches ovule through micropyle

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3. Two male gametes (n) are released

Fertilisation:

- First fertilisation: 1 male gamete + egg cell (n) → Zygote (2n) → develops into Embryo
- Second fertilisation: 1 male gamete + 2 polar nuclei → Primary Endosperm Nucleus (3n) → Endosperm (food for embryo)

This is called double fertilisation (syngamy + triple fusion).

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Q6. What is Polymerase Chain Reaction (PCR)? State its uses.

[2 marks]

PCR is an in vitro technique used to amplify specific segments of DNA millions of times.

Steps (repeated 30-40 cycles):

1. Denaturation: 94°C - DNA strands separate
2. Annealing: 54-60°C - primers attach to complementary sequences
3. Extension: 72°C - Taq polymerase synthesises new DNA strands

Uses:

- Crime investigation (DNA fingerprinting)
- Diagnosis of genetic diseases
- Detection of HIV, COVID-19
- Paternity testing, evolutionary studies
- Cloning and gene expression studies

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Q7. Differentiate between innate and acquired immunity.

[2 marks]

INNATE IMMUNITY:

- Present from birth, non-specific
- First line of defence
- Does not improve with repeated exposure
- Examples: skin, mucus, fever, phagocytes, NK cells, lysozyme

ACQUIRED IMMUNITY:

- Developed after exposure to pathogen or vaccine
- Specific to particular pathogen
- Has memory (faster response on 2nd exposure)
- Types: Humoral (B cells, antibodies) and Cell-mediated (T cells)
- Examples: antibodies against measles after vaccination

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SECTION C — Long Answer (5 marks each)

Q8. Explain the process of DNA replication in prokaryotes. Name the enzymes involved.

[5 marks]

DNA REPLICATION IN PROKARYOTES (E. coli):

Semi-conservative replication: Each new DNA has one original + one new strand.

(Proved by Meselson and Stahl experiment using N-15 and N-14)

Process:

1. INITIATION:

- Begins at ori C (origin of replication)
- DnaA protein unwinds DNA at AT-rich region
- DNA is circular in prokaryotes

2. UNWINDING:

- Helicase enzyme breaks hydrogen bonds between strands

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- Single Strand Binding proteins (SSB) stabilise single strands
- Topoisomerase (Gyrase) prevents supercoiling ahead of fork
- Two replication forks move in opposite directions (bidirectional)

3. PRIMER SYNTHESIS:

- Primase synthesises short RNA primer (5-10 nucleotides)
- Provides a free 3'-OH group for DNA polymerase

4. ELONGATION:

- DNA Pol III adds dNTPs in 5'→3' direction only
- Leading strand: continuous synthesis
- Lagging strand: discontinuous (Okazaki fragments)

5. PRIMER REMOVAL AND JOINING:

- RNase H removes RNA primers
- DNA Pol I fills gaps with DNA
- DNA Ligase joins Okazaki fragments

KEY ENZYMES:

Helicase, Primase, DNA Pol I & III, Topoisomerase, Ligase, SSB proteins

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Q9. What is biotechnology? Describe the tools and techniques used in recombinant DNA technology.

[5 marks]

BIOTECHNOLOGY: Application of biological organisms or their products for industrial, medical, or agricultural purposes.

Modern biotechnology includes Recombinant DNA technology, cell and tissue culture.

TOOLS OF RECOMBINANT DNA TECHNOLOGY:**1. RESTRICTION ENZYMES (Molecular Scissors):**

- Cut DNA at specific palindromic sequences (4-8 bp)
- EcoRI cuts at 5'-GAATTC-3'

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- Produce sticky ends (staggered cuts) or blunt ends
- First restriction enzyme discovered by Werner Arber

2. DNA LIGASE (Molecular Glue):

- Joins cut DNA fragments together
- Creates phosphodiester bonds between sugar-phosphate backbone
- Used to insert foreign DNA into vector

3. VECTORS:

- Carry foreign DNA into host cell
- Types: Plasmids (pBR322), Bacteriophages (λ phage), Cosmids, BAC, YAC
- Must have: ori, selectable marker, cloning site

4. HOST ORGANISMS:

- E. coli (most common), Agrobacterium, Bacillus, yeast
- Allow replication and expression of recombinant DNA

5. PCR:

- Amplifies specific DNA fragments in vitro
- Used to produce large quantities of target DNA

6. GEL ELECTROPHORESIS:

- Separates DNA fragments by size in agarose gel
- Smaller fragments move faster toward positive electrode
- Visualised under UV with ethidium bromide staining

PROCESS: Cut DNA with restriction enzyme + Ligate into vector + Transform into host + Select recombinants

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*** END OF ANSWER BOOKLET ***

MARKS SUMMARY (For Examiner Use Only)

Section	Max Marks	Marks Obtained	Examiner Initials
Section A (1 mark × 4)	4		
Section B (2 marks × 3)	6		
Section C (4-5 marks)	varies		
Section D (5-6 marks)	varies		
TOTAL	100		